Physics-Informed Machine Learning, with No Data

There are no measurements: the laws of physics are your data, and your model must obey them

Physics-informed ML **Data you create from equations** **Uses Weeks 7, 8 and 11**

The task. An engineer needs a fast, reliable model of a cooling system: how hot a part gets when its power and airflow change. Nobody has measured it. You will use the physics equations to create data, train machine-learning models on it, and show that a model which builds in the physics predicts better outside the data it was trained on.

THE DATA

There is no dataset. In Week 9, each student gets their own version of the system, with its own numbers and one unknown setting, worked out from their student number. So no two projects have the same answer, and no AI assistant has seen yours. You write a small program, a simulator, that solves the equation step by step, and use it to create clean data and data with added noise.

WHAT YOU WILL DO

1. Write the simulator and check it against a simple case where the exact answer is known.
2. Train an ordinary neural network on the simulated data, and test it both inside and outside the range it was trained on.
3. Build a physics-informed model: one that keeps the known physics and learns only the missing part, a neural ODE (Week 11), or one penalised for breaking a law of physics. Compare how well it predicts outside its training range.
4. Work backwards: find your unknown setting from noisy simulated measurements, and say how sure you are.

WHAT YOU CODE YOURSELF

Write **backpropagation** and an **optimiser step** for your networks and check them against PyTorch. Use **PCA via SVD** on many simulated runs to find how few patterns describe the system; engineers call this proper orthogonal decomposition. Check it against NumPy.

IS IT FAIR AND SAFE?

There are no people in this data, so fairness becomes **reliability**: does the model work equally well in every situation? Measure its error at low and high power, over short and long periods, and with clean and noisy inputs. Improve the weakest situation with one change and measure the cost elsewhere. Compare each learned physical setting with its true value. Write down exactly where the model may be used, and what testing it would need before an engineer could rely on it.

QUESTIONS TO PREPARE FOR

Why does the physics-informed model predict better outside its training range, and where does it still fail? How do you know your simulator is right? What would you do if real measurements arrived next week?

COMMON MISTAKES

Only testing inside the training range. A simulator that takes steps too large to be accurate. Saying you found the unknown setting without saying how sure you are.

WEEK-BY-WEEK PLAN

Week 9: your version received; simulator written and checked; clean and noisy data created. **Week 10:** ordinary and physics-informed models trained and compared; hand-written pieces passing their tests. **Week 11:** unknown setting found, with how sure you are; reliability check; PCA via SVD. **By 6 January 2027:** report written; repository checked and submitted. **Week of 11 January:** 10-minute in-person interview.

TO AIM HIGHER

Compare a neural ODE and a hybrid model on the same test. Give a range for the unknown setting by repeating the fit on many noisy datasets. Show how accuracy drops as the noise grows. Check that your model conserves the energy the physics says it must.